

Title: Comprehensive Validation of Einthoven's and Goldberger's Electrocardiographic Lead Systems Using the PTB-XL Dataset

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HRS Abstract Link:

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Background

Einthoven's and Goldberger's lead systems are the basis of electrocardiography, providing insights into cardiac function and heart health. Despite their long-standing use, the new landscape of technology and lack of current validation of these systems require a re-evaluation.

Objective

This study aims to validate Einthoven's and Goldberger's lead systems and their computational accuracy compared to recorded leads. It focuses on assessing these leads' accuracy, diagnostic utility, and reliability in detecting various cardiac diseases. Additionally, this study aims to explore the integration of these lead systems with new emerging technologies, such as wearable ECG devices.

Methods

The primary focus was on evaluating the accuracy of Einthoven's and Goldberger's formulas by comparing the calculated lead values with the actual recorded values in the dataset. The study also explored how diseases impacted the accuracy of the computed leads.

Results

The analysis indicated that Einthoven's and Goldberger's formulas generally maintain high accuracy even with abnormal ECGs. Each lead had an average error of less than 0.3%, validating these lead systems again. Average Errors: Lead II: 0.00115%, Lead aVF: 0.24245%, Lead aVL: 0.01137%, Lead aVR: 0.01149. However, certain variations were observed, particularly with inaccurately recorded ECGs, resulting in a mismatch of computed and actual readings. One limitation of this study arises from the variability in how different electrocardiogram machines record and compute leads. Different machines record Lead 2 while others compute it, adding a layer of uncertainty.

Conclusion

The study confirms the computational accuracy of Einthoven's and Goldberger's lead systems. They maintain renewed importance for emerging technologies such as wearable devices, which can record only a limited selection of leads, necessitating the computation or synthesis of additional leads. This study provided a contemporary re-validation of these systems where there

was a significant gap in research. This study also supports the ongoing use of these traditional lead systems in clinical practice while highlighting areas for future research in the field of Artificial Intelligence and ECGs.